



**CS301- DATA STRUCTURE
(SOLVED MCQs)
FROM MIDTERM PAPERS
LECTURE (1-22)**



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CS619 & CS519

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1. Each node in singly link list has

✚ 1 pointer

✚ 2 pointers

✚ 3 pointers

✚ 4 pointers

2. Parameters in function call are passed using,

✚ Stack

✚ Queue

✚ Binary Search Tree

✚ AVL Tree

3. linked list is always contains elements that can be described as;

✚ Redundant

✚ Recursive

✚ Self-referential

✚ Bidirectional

4. The..... method of list data structure removes the element residing at the current position.

✚ Add

✚ Next

✚ Remove

✚ Find

5. One difference between a queue and a stack is:

✚ Queues require dynamic memory, but stacks do not

✚ Stacks require dynamic memory, but queues do not

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✚ Queues use two ends of the structure; stacks use only one

✚ Stacks use two ends of the structure; queues use only one

6. Insertion in a linked list can be done at,

✚ Front only

✚ Back only

✚ Somewhere in middle only

✚ Front, back and somewhere in the middle

7. An array is a group of.....memory locations.

✚ Scattered

✚ Isolated

✚ Random(non-consecutive)

✚ Consecutive

8. Which of the following applications may use a stack?

✚ Accessing shared resource

✚ Parentheses balancing program

✚ Buffering messages

✚ Waiting list

9. Every AVL is.....

✚ Ternary Tree

✚ Complete Binary Tree

✚ Heap

✚ Binary Search Tree

10. Consider the following infix expression:

$$x - y * a + b / c$$

Which of the following is a correct equivalent expression for the above?

✚ $x y - a * b + c /$

✚ $x * y a - b c / +$

✚ $x y a * - b c / +$

✚ $x y a * - b / + c$

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11. Consider the following infix expression. $7/8+9$

If one converts the above expression into postfix, what would be the resultant expression?

- ☐ 789/+
- ☐ 78/+9
- ☐ /78+9
- ☒ 78/9+

12. Suppose there are 100 elements in an equivalence classes so initially there will be 100 tree. The collection of tree is called.....

- ☐ Cluster
- ☐ Class
- ☒ Forest
- ☐ Bunch

13. For a perfect binary tree of height h , having N nodes the sum of height of nodes is

- ☒ $N-h-1$
- ☐ $N-1$
- ☐ $N-1+h$
- ☐ $-(h-1)$

14. Sorting procedure normally takes.....time

- ☒ $N \log N$
- ☐ $2N$
- ☐ $N*N*N$
- ☐ N

15. there are 100 elements in an equivalence class then will havestate initially

- ☐ 50
- ☒ 100
- ☐ 1000
- ☐ 80

16. objects (objects accessed by pointers) are called anonymous objects.

- ☐ Private
- ☒ Nameless
- ☐ Friend
- ☐ Public

17. is a self-balancing tree.

- ☒ AVL
- ☐ Binary Tree
- ☐ Binary Search Tree
- ☐ ALV

18. Which of the following operation returns but do not removes top value of the stack?

- ☐ Push
- ☐ Pop
- ☒ Top
- ☐ First

19. `Int htdiff = height(root->getLeft())..... height(root->getRight());` The above line of code is taken from AVL insert method.

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Complete it by selecting an appropriate symbol.



+

/

*

20. Which operation of the queue data structure is used to insert an element into the Queue?

Enqueue()

Dequeue()

Fornt()

Remove()

21. For making binary search tree for strings we need, data type.

Char

Int

Float

Double

22. Local variables defined inside function body areautomatically at the end of function execution.

Created

Destroyed

Incremented

Decrementd

23. If a node has three fields, then this node will be related to

Linked list

Doubly linked list

Circular linked list

All of the given

24.is an area in computer memory that is allocated dynamically.

Heap

Stack

Queue

Linked list

25. Linked list use.....to store data.

Array

2-D Array

Variables

Linked Memory

26. The lifetime of a transient object cannot exceed that of the application.

True

False

In some cases

None of the given

27. To represent hierarchical relationship between elements, which data structure is suitable?

Dequeue

Priority

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✚ Stack

✚ Tree

28. Local variables of a function are stored in,

✚ Binary search tree

✚ Stack

✚ Queue

✚ AVL tree

29. In-order traversal method traverses the data in

✚ Non sorted order

✚ Random order

✚ Sorted order

✚ None of the given

30. Dequeue() operation of queue data structure is used to.....

✚ Get an element from the front of the queue

✚ Remove an element from the front and return it

✚ Insert an element at the front

✚ Insert an element at the back

31. In a tree, we link the nodes in such a way that it.....a linear structure.

✚ Does not remain

✚ Forms

✚ Reverses

✚ Remains

32. Factorial is an example of.....function.

✚ Recursive

✚ Non-recursive

✚ Cube

✚ Log

33. All the objects created using.....operator have to be explicitly destroyed using delete operator.

✚ New

✚ Delete

✚ Del

✚ Create

34. In the calling function, after the execution of the function called, the program continues its execution from the.....after the function call.

✚ Previous line

✚ Next line

✚ Beginning

✚ None of the above

35. Last node in circular linked list contains

✚ At least one null pointer

✚ No null pointer

✚ Maximum two null pointers

✚ None of the given

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36. One should be careful about transient.....that are stored by reference in data structures.

☒ Objects

☐ Stack

☐ Function

☐ Tree

37. Binary search tree violates the condition of AVL tree when any node has balance equal to

☐ 2 or -2

☒ 1 or -1

☐ 0

☐ None of the options

38. Security of data is the main usage of AVL tree.

☐ True

☒ False

☐ In some cases

☐ None of the given

39. A binary tree whose every node has either zero or two children is called _____.

☒ Complete binary tree

☐ Binary search tree

☐ Strictly binary tree

☐ None of above

40. In internal memory organization of a process, there is some area of memory for static data that holds.....variables.

a. Static

b. Global

c. Not static neither global

☒ d. Both static and global

41. Following is true in case of using recursive method calls.

☐ The code becomes very long

☐ There is no effect on length of code

☒ The code becomes very short

☐ Code becomes very easy to understand

42. In which traversal method, the recursive calls can be used to traverse a binary tree?

☐ In preorder traversal only

☐ In inorder traversal only

☐ In postorder traversal only

☒ All of the given options

43. If we use doubly linked list to implement list then there is an issue of

☐ Next pointer of first node and pre pointer of last node are NULL

☐ Next pointer of first node and next pointer of last node are NULL

☒ Pre pointer of first node and next pointer of last node are NULL

☐ Pre pointer of first node and pre pointer of last node are NULL

44. In case of insertion of left outer node in BST,

☐ We apply single right rotation to make it AVL tree.

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- ☐ We apply single left rotation to make it AVL tree.
 - ☐ We first apply right rotation and then left rotation to make it AVL tree.
 - ☒ We first apply left rotation and then right rotation to make it AVL tree.
45. Whenever we call a function, the compiler makes a stack, the top element of the stack is.....
of the function.
- ☐ First argument
 - ☒ Return address
 - ☐ Last argument
 - ☐ None of the above
46. Suppose a stack class has been defined using template. Now, we want to declare a Stack object of an int type. What will be the correct syntax?
- ☐ <int>Stack stack;
 - ☒ Stack<int> stack;
 - ☐ Stack int stack;
 - ☐ Int Stack stack;
47. In node class one field is an integer data and other field will be.....
- ☐ Pointer to class
 - ☒ Pointer to node
 - ☐ Pointer to integer
 - d. None of given options
48. If there are 100,0000 unique members (nodes) stored in a complete binary tree, the tree will have.....levels.
- ☐ 10
 - ☒ 20
 - ☐ 30
 - ☐ 40
49. ~Binary Search Tree () is a.....
- ☐ Constructor
 - ☒ Destructor
 - ☐ Switch case
 - ☐ Template method call
50.is utilized at the time of memory allocation in dynamic manner.
- ☐ Stack
 - ☐ Queue
 - ☒ Heap
 - ☐ All of the given
51. How many cases of rotation are there in AVL tree?
- ☐ 2
 - ☒ 4
 - ☐ 6
 - ☐ 8
52. A list is the collection of items of the.....
- ☒ May be of same or may be of different type

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- ☐ Different type
- ☒ Same type
- ☐ None of the above

53. `Int * i = new
int[10];` Above
given code will:

- ☐ Create an integer having value 10
- ☐ Allocate memory for 9 integers
- ☒ Allocate memory for 10 integers
- ☐ Create 10 pointers of integer type

54. The depth of a binary tree is

- ☐ Total number of nodes in the tree
- ☐ Number of leaf nodes in the tree
- ☐ Number of non-leaf nodes in the tree
- ☒ Maximum level of a leaf

55. A.....model attempts to model a real-world phenomenon

- ☐ Physical
- ☐ Logical
- ☒ Simulation
- ☐ Conceptual

56. In level-order traversal for binary search tree, we visit the nodes at each level before proceeding to the next level, in a.....order.

- ☐ Right-to-left
- ☒ Left-to-right
- ☐ Top-to-bottom
- ☐ Bottom-to-top

57. The balance of a node in a binary tree is defined as the height of its.....sub tree minus height of its right sub tree.

- ☐ Right
- ☒ Left
- ☐ Upper
- ☐ Lower

58. Allocating and de-allocating memory for linked list nodes does take.....time than pre-allocated array.

- ☐ Less
- ☒ More
- ☐ Equal
- ☐ No

59. Consider we have performed the following operations on a stack of size 5. `Push(10);`
`Push(20);`
`Push(30);`
`Pop();`

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Pop();
Pop();
Push(40);
Push(50);
Pop();

After the completion of all operation, the top element in stack is _____.

- ☐ 10
- ☐ 20
- ☒ 40
- ☐ 50

60. In doubly linked list a node consists of three parts:

- ☐ 1 pointer and 2 objects
- ☒ 2 pointers and 1 object
- ☐ 3 objects
- ☐ 3 pointers

61. For reference variables, _____ sign is used.

- ☒ a. Ampersand
- ☐ b. Asterisk
- ☐ c. Sigma
- ☐ d. Dollar

62. Which of the following data structure is linear type?

- ☐ Stack
- ☐ List
- ☐ Queue
- ☒ All of the above

63. The -- is a decrement operator in C++ that decreases the value of the operand by _____.

- ☒ One
- ☐ Two
- ☐ Three
- ☐ Four

64. Which of the following statement is correct for the variable "current- -"?

- ☐ Current = current + 1
- ☒ Current = current - 1
- ☐ Current = current - 2
- ☐ Current - 1 = current

65. When an executable program runs, it is loaded in the computer memory and becomes a _____.

- ☒ a. Thread
- ☐ b. .h file
- ☒ c. Process
- ☐ d. None of the above

66. start() method of list class is used to:

- ☒ Moves the "current" pointer to very first element
- ☐ Moves the "current" pointer to very last element

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- ☒ Moves the “current” pointer to one step after the first element of the array
- ☒ Moves the “current” pointer to one step before the first element of the array

67. $a * (b + c) - d$ is an example of _____ expression.

- ☒ Infix
- ☐ Prefix
- ☐ Postfix
- ☐ Alfix

68. In a program a reference variable, say x, can be declared as

- ☒ `int &x;`
- ☐ `int *x;`
- ☐ `int x;`
- ☐ none of the given options

69. We allocate memory dynamically by using _____ operator.

- ☐ This
- ☒ New
- ☐ Increment

d. Decrement

70. Get(?) method of list class is used to:

- ☐ Get element from the last position
- ☐ Get element from the first position
- ☐ Get element from the middle position
- ☒ Get element at the given position

71. STL is a _____ that is a part of the official standard of C++.

- ☒ C program file

b. .h file

c. .cpp file

d. Library

72. There are _____ cases for deleting a node from binary search tree.

- ☐ 1
- ☐ 2
- ☒ 3
- ☐ 4

73. In a list, tail() method of current pointer _____

- ☐ Returns the last element of the “current” pointer
- ☐ Moves the “current” pointer to the very first element
- ☒ Moves the “current” pointer to the very last element
- ☐ Returns the first element of the “current” pointer

74. Suppose we have been given the following data set for a queue:

37524

What will be the resultant queue if we call a front() method?

- ☐ 7524
- ☒ 37524

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 75243

 5 2 4

75. Suppose we have been given the following data set for a queue:

7524

What will be the resultant queue if we call enqueue(3) method?

Note that 7 is the front element whereas 4 is rear element of queue.

 7524

 37524

 75243

 5243

76. Maximum time that an insertion operation can take in AVL tree is _____. Here log stands for log to the base of 2.

 Log (n)

 1.44 log (n)

 1.66 log (n)

 Log (n+1)

77. _____ is when function is calling to itself.

 Loop

b. Recursion

c. Iteration

d. Nested loop

78. Left, right, info, and parent are the operations of _____ data structure.

 Stack

 Tree

 Queue

 Linked list

79. Following is a keyword of C++

 Del

 Delete

 Remove

 Eliminate

80. The expression DE+H* is called _____

 Prefix expression

 Infix expression

 Postfix expression

 Hybrid expression

81. A software solution is said to be efficient if it solves the problem _____.

 By using some extra resources

 Within no time

 By consuming more hardware resources

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✚ Within its resources constraints

82. Each node in a singly linked list contains two fields, one field called data field while other field contains:

- ✚ Pointer to an integer
- ✚ Pointer to character
- ✚ Pointer to next node
- ✚ Pointer to class

83. Each node in singly linked list contains _____

- ✚ One pointer
- ✚ Two pointers
- ✚ No pointer
- ✚ NULL pointer

84. Every AVL tree is a binary search tree.

- ✚ True
- ✚ False
- ✚ Not in some cases
- ✚ None of the given

85. In singly linked list “next” field of node contains:

- ✚ Address of next node
- ✚ Object of next node
- ✚ Object of current node
- ✚ Address of head node

86. Two common models of simulation are _____ and _____.

- ✚ Circuit-based simulation and Event- based simulation
- ✚ Circuit-based simulation and Time-based simulation
- ✚ Time-based simulation and Event- based simulation
- ✚ None of the above

87. What are the basic things associated with data structures?

- ✚ Space for each data item it stores
- ✚ Time to perform each basic operation
- ✚ Programming effort
- ✚ All of the above

88. In AVL tree insertion occurs on the inside in case _____ and 3 which a single rotation cannot fix.

- ✚ 1
- ✚ 2
- ✚ 4
- ✚ 5

89. add(12) method of linked list class will

- ✚ Add 12 nodes in linked list
- ✚ Add 12 pointers in linked list
- ✚ Add 12 as value in linked list
- ✚ Add 12 values in linked list

90. Which of the following is a non- linear data structure?

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- ✚ Stack
- ✚ Queue
- ✚ Tree
- ✚ Linked list

91. Which data structure allows deleting data elements from front and inserting at rear?

- ✚ Stacks
- ✚ Queues
- ✚ Deques
- ✚ Binary search tree

92. Array cells are ____ in computer memory.

- ✚ Contiguous
- ✚ Random
- ✚ Store in multiple Variables
- ✚ Store in multiple functions

93. A kind of expression where the operator is present between two operands called ____ expressions.

- ✚ Infix
- ✚ Postfix

c. Prefix

d. None of the above

94. back() method of list class is used to

- ✚ Moves the "current" pointer to backward one element.
- ✚ Moves the "current" pointer to backward two element.
- ✚ Moves the "current" pointer to backward three element.
- ✚ Moves the "current" pointer to backward four element.

95. next() method of List class is used to:

- ✚ Moves the Current position backward one element
- ✚ Moves the "Current" pointer to two steps after the last element of the array
- ✚ Moves the Current position forward one element
- ✚ Moves the "Current" pointer to two steps before the last element of the array

96. ____ tree has been named after two persons Adelson-Velskii and Landis.

- ✚ Binary
- ✚ Black
- ✚ AVL
- ✚ VLA

97. There are four cases of rotation in an ____ tree.

- ✚ ELV
- ✚ EVL
- ✚ AVL
- ✚ ALV

98. ____ is used for reference variable in C++.

- ✚ !
- ✚ @

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#

&

99. A _____ is a tree in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible.

✚ Strict binary tree

✚ Full binary tree

✚ Perfect binary tree

✚ Complete binary tree

100. A queue is a data structure where elements are

✚ Inserted at the front and removed from the back

✚ Inserted and removed from the top

✚ Inserted at the back and removed from the front

✚ Inserted and removed from both ends

101. Longest path from root node to farthest leaf node is called _____ of tree.

✚ Level

✚ Length

✚ Depth

✚ Node level

102. New items are added at the _____ of the stack.

a. Bottom

b. Middle

c. Center

d. Top

103. Length() method of list class is used to:

✚ Return the length of the array

✚ Return the length of the list

✚ Return the length of empty part of the array

✚ Return the length of empty part of the list

104. The function calls are made with the help of _____.

✚ Stack

✚ Heap

✚ Dynamic memory

✚ External memory

105. An efficient program execute faster and helps in _____ the usage of resources like memory and disk

✚ Maximizing

✚ Minimizing

✚ Equalizing

✚ None of the given

106. Generalized code written for a class is called:

✚ Function

✚ Template

✚ Structure

✚ Stack

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107. "set()" method of list class is used to:
- ☐ Set the value of Pointer
 - ☒ Set the value of Null Nodes
 - ☐ Set the value of objects
 - ☐ Set the value of Value
108. In C++, we place the class interface in ____ file.
- .cpp
 - .cppp
 - .h**
 - .hh
109. Every ____ tree is a binary search tree.
- ☒ AVL
 - ☐ binary
 - ☐ big
 - ☐ small
110. If both left and right nodes of a node are NULL then this type of node is called ____ node.
- ☐ Non leaf
 - ☐ internal
 - ☐ inner
 - d. leaf**
111. Which of the following function don't belongs to the stack class?
- ☐ push()
 - ☐ pop()
 - ☒ crash()
 - ☐ top()
112. For a complete binary tree, the depth is calculated as ____
- ☒ $\log_2(\text{number of nodes}+1)-1$
 - ☐ $\log_2(\text{number of nodes}+1)+1$
 - ☐ $\log_2(\text{number of nodes}-1)-1$
 - ☐ $\log_2(\text{number of nodes}-1)+1$
113. ____ only removes items in reverse order as they were entered
- ☐ Queue
 - ☒ Stack
 - ☐ Both of these
 - ☐ None of these
114. What will be the postfix expression of following infix expression? $D+E*F/G$
- ☐ $DE*F/G$
 - ☐ $DE+F*G/$
 - ☐ $DEF*/+$
 - ☒ $DE+FG*/$
115. The ____ of a node in a binary tree is defined as the height of its ____ sub tree minus height of its right sub tree.
- ☐ Height

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- ☒ Balance
 - ☐ Width
 - ☐ None
116. The back() method decreases the value of Variable current by ____
- ☐ Four
 - ☐ Three
 - ☐ Two
 - ☒ One
117. In doubly linked list there is/are.....pointer/s in each node.
- ☐ One
 - ☒ Two
 - ☐ Three
 - ☐ Four
118. Suppose we have the following values to be inserted in constructing AVL tree, 10, 13, 15, 5, 7, 8
Tell when first rotation will take place.
- a. ☒ After inserting node 13
 - b. ☐ After inserting node 15
 - c. ☐ After inserting node 5
 - d. ☐ After inserting node 7
119. Binary search algorithm cannot be applied to ____
- ☒ Sorted linked list
 - ☐ sorted binary trees
 - ☐ sorted linear array
 - ☐ None of given option
120. During the execution of a process operating system constructs focus things for that process which of the following is not part of that process
- ☐ A section for static data including global variable
 - ☐ Stack
 - ☒ Heap
 - ☐ Linked list
121. Which one of the following method does not change the original value of the argument in the calling function?
- ☐ Call by passing reference of the argument
 - ☐ Call by passing the address of the argument
 - ☒ Call by passing the value of the argument
 - ☐ None of the given options
122. ____ is the major factor to see the efficiency of a program.
- ☐ Quality
 - ☒ Time
 - ☐ Correctness
 - ☐ None of the given
123. While implementing stack with an array and to achieve LIFO behavior, we used push and pop elements at ____.

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- ☐ The start of the array
 - ☒ The end of the array
 - ☐ The mid of the array
 - ☐ At least one position before array starting index.
124. A template is a function or class that is written with a _____.
☐ Specific
☐ Definite
☒ Generic
☐ None of the above
125. In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?
☐ After all other entries that are greater than the new entry
☒ At the head
☐ After all other entries that are smaller than the new entry
☐ At the tail
126. The next field in the last node in a singly-linked list is set to _____.
☐ 0
☐ 1
☒ NULL
☐ False
127. Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While current pointer is pointing memory location having 72 stored in it. After calling remove() function on the following linked list current point will point to memory location having value?
☒ 6
☐ 35
☐ 65
☐ 25
128. If we write functions for recursive and non recursive inorder traversal method of BST, what will be the difference between its functions prototypes?
☐ Different return types
☐ Different function names
☐ Different arguments list
☒ Nothing will be different
129. In singly linked list a node comprises of _____ field/s.
☐ One
☒ Two
☐ Three
☐ Four
130. Doubly linked list always has _____ NULL pointers in a node.
☐ One
☒ Two
☐ Three

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- ✚ Four
131. A BST generated from the data in ascending order is _____.
✚ Linear
✚ Nonlinear
✚ Balanced
✚ Un sorted
132. Linked list is generally considered an example of _____ type of memory location.
✚ Static
✚ Compile time
✚ Dynamic
✚ None of given options
133. What's wrong with following loop?
`while((i < 10) && (i > 24)){`
`}`
a. The logical operator && cannot be used in a test condition
b. The while loop is an exit-condition loop
✚ c. The test condition is always false
d. The test condition is always true
134. Making the tree unbalanced, it violates the _____ rule.
✚ Linked list
✚ Stack
✚ AVL
✚ Queue
135. copy() method of list data structure _____.
✚ copy first item of list
✚ set one list to be a copy of another
✚ copy last item of list
✚ copy any item of list
136. Deleting a _____ node in BST is a _____ case.
✚ Root, simplest
✚ Left child, simplest
✚ Right child, simplest
✚ Leaf, simplest
137. We can not remove items randomly from _____.
✚ Stack
✚ Queue
✚ Both of these
✚ None of these
138. NULL is an invalid address and _____.
✚ Accessible
✚ Inaccessible
✚ Points to the start point of the list
✚ Points to the last point of the list

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139. A stack carries _____ behavior.
- ☐ FIFO
 - ☒ LIFO
 - ☐ AVCO
 - ☐ FEFO
140. The tree data structure is a
- ☐ Linear data structure
 - ☒ Non-linear data structure
 - ☐ Graphical data structure
 - ☐ Data structure like queue
141. Can we store elements with different data types in a single array?
- ☐ Yes
 - ☒ No
 - ☐ In some cases
 - ☐ None of given
142. The lifetime of a transient object can exceed that of application which is accessing it.
- ☐ True
 - ☒ False
 - ☐ In some cases
 - ☐ None of the given
143. In a complete binary tree, for 25000 nodes the depth will be _____.
- ☐ 13
 - ☒ 14
- c. 15
d. 16
144. The smallest value element in a binary search tree(Each node with left and right pointer) lies at
- ☐ Root Node
 - ☐ Left Child of Root
 - ☐ Right Most Node
 - ☒ Left Most Node
145. if we use array to implement list, then there is an issue that is gives difficulty when
- ☐ We will access values randomly
 - ☐ We will remove data from it
 - ☐ We will increase its size
 - ☐ We will decrease its size
146. if there are ____ nodes in an avl tree its levels will be roughly as $\log_2(10 \text{ million})$
- ☐ 100 million
 - ☒ 10 million
 - ☐ 5 million

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2 million

147. Which one is the correct function call for the following function of calculatincube?

```
int cube(int&num)
{
.
.
.
}
```

- a. Cube(&num)
- b. Cube(&&num)
- c. Cube(*num)
- d. Cube(num)

148. Which of the following is the correct option for priority Queue?

- The type of Queue that is not FIFO i.e the person who comes first may not leave first
- The type of Queue that is not FIFO i.e the person who comes first should leave first
- The type of Queue that is not FIFO i.e the person who comes first should leave first
- The type of Queue that is not FIFO i.e the person who comes first may not leave first

149. For String-based Binary Search Tree, We use ASCII values of characters for comparing among letters. This method is known as ____

- Lexicographic order
- Alphabet coding procedure
- Asymmetric technique
- heap-based approach

150. Elements in a queue data structure are added from ____ and remove from ____ .

- Rear end. front end
- Front end. Rear end
- Front end
- rear end

151. If numbers 5,222,4,48 are inserted in a queue. Which one will be remove first?

- i. 48
- ii. 4
- iii. 222
- iv. 5

152. A zigzag rotation is performed. In Left-Left case of rotation in AVL tree.

- i. True
- ii. False
- iii. In some cases
- iv. None of the above

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153. During deletion of node from BST. if we found this node don't have in- order successor and predecessor. it means this node is ____
- i. Left Most node in the binary search tree
 - ii. Right most node in the binary search tree
 - iii. Root node
 - iv. None of the given option
154. ____ rule applies for evaluating operators of same precedence in an expression
- i. right to left
 - ii. Cascading
 - iii. Associative
 - iv. None of the above
155. What will be result of following postfix expression?
- 23*+2-
- i. 3
 - ii. 4
 - iii. 5
 - iv. 10
156. $y = \&x[0];$
In the above statement, we get the address of the first location of the array x and store it in y.
Here “y” is:
- a. rvalue
 - b. xvalue
 - c. lvalue
 - d. zvalue
157. what will be postfix expression of the following infix expression? Infix expression $a+b*c-d$
- a. $ab+c*d-$
 - b. $abc*+d-$
 - c. $abcd+*-$
 - d. $abc+*d-$
158. Which of the following is the correct conversion of infix to postfix expression? $Z+B-(D-H)/K$
- a. $ZB+D-H-K/$
 - b. $ZB+DH-K-/$
 - c. $ZB+DH-K/-$
 - d. $ZB+DHK--/$
159. Leaf node of binary search tree contains.....
- a. One null pointer
 - b. Three null pointers
 - c. Two null pointers
 - d. All of the given
160. A tree is an AVL tree if
- a. Any one node fulfills the AVL condition

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- ☐ At least half of the nodes fulfill the AVL condition
 - ☐ All the nodes fulfill the AVL condition
 - ☐ None of the given options
161. The.....of a binary tree is the maximum level of its leaves (also called the depth).
- ☐ Level
 - ☐ Width
 - ☒ Height
 - ☐ None of the above
162. Memory address is stored in
- ☐ Address operator
 - ☐ Reference
 - ☒ Pointer
 - ☐ All of the given
163. A binary tree is said to be a.....binary tree if every non-leaf node in a binary tree has non-empty left and right sub trees.
- ☐ Complete
 - ☒ Strictly
 - ☐ AVL
 - ☐ Perfect
164. “+” is a.....operator.
- ☐ Unary
 - ☒ Binary
 - ☐ Ternary
 - ☐ None of the above
165. Which of the following a nonlinear tree
- a. Stack
 - b. Que
 - ☒ c. Tree
 - d. Linked List
166. For every Process execute the last part of the Memory is for.....of the Program
- a. Data
 - b. Code
 - c. Stack
 - ☒ d. Heap
167. While implement non recursive traversal for binary search tree we need to implement_____
- a. Que
 - ☒ b. Stack
 - c. Min heap
 - d. Max Heap
168. Suppose we have the following values to be inserted in constructing AVL tree, 20,23,25,10, 12,13

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Tell when first rotation will take place.

a. After increasing Node 25

b. After increasing Node 23

c. After increasing Node 10

d. After increasing Node 12

169. If a tree has a 50 Node then the total linked in the tree will be

a. 55

b. 51

c. 50

d. 49

170. The percolate down procedure will move the smaller value ____ and Bigger value ____

a. Left, right

b. Right, left

c. Down, up

d. Up, down

171. Circular Linked List Solve the problem of pointer/method of the doubly link list.

a) Remove

b) Null

c) Add

d) Find

172. isEmpty() method of stack class will return true when.

a) Stack is full

b) Stack is Partially

c) Stack is not empty or Null

d) Stack is empty

173. "New int[11]" will allocate memory for integers.

a) 13

b) 12

c) 11

d) 10

174. For searching a particular number in Binary Search Tree (if it is not present), the maximum number of comparisons will be _____ comparison at each level.

a) 1

b) 2

c) 3

d) 4

175. In singly linked list which node will keep track of starting position of the list.

a) Next Node

b) Previous Node

c) Head Node

d) Last Node

176. If we singly linked list to implement list, then there is an issue that it gives difficulty when we:

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- a) Move forward in the list
 - b) Move backward in the list
 - c) We will increase its size
 - d) We will decrease its size
177. Can we store elements with different data types in a single array?
- a) Yes
 - b) No
 - c) In some cases
 - d) None of the given
178. During union by size method, which data structure is used to improve the balancing of tree?
- a) Array
 - b) Stack
 - c) Linked List
 - d) Tree
179. The difference between a "Binary Tree (BT)" and a "Binary Search Tree (BST)" is Which that,
- a) A BST has two children per node whereas a BT can have none, one or two children per node
 - b) In BST nodes are inserted based on the values they contain
 - c) In BT nodes are inserted based on the values they contain
 - d) There is no difference
180. The main use of AVL tree is:
- a) Searching of data
 - b) Storing of data
 - c) Insertion of data
 - d) Security of data
 - e) The expression
181. if (! heap ->isEmpty())
checks
- a) Heap is empty
 - b) Heap is full
 - c) Heap is not empty
 - d) Not a valid expression
182. A solution is said to be efficient if it solves the problem within its resource constraints i.e. hardware and time.
- a) True
 - b) False
183. A queue where the dequeue operation does not depend upon FIFO, is called:
- a) enqueue
 - b) simple queue
 - c) stack
 - d) priority queue
184. In simple or singly linked list there is/are.....pointer/s in each node.
- a) One
 - b) Two

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- c) Three
d) Four
185. Which one of the following statement is correct?
a) Array size is fixed once it is created
b) Link List size is fixed once it is created
c) Binary Search Tree size is fixed once it is created
d) AVL Tree size is fixed once it is created
186. Which of the is NOT true regarding the maze generation?
a) Randomly remove walls until the entrance and exit cells are in same set.
b) Removing a wall is the same as doing a union operation
c) Remove a randomly chosen wall if the cells it separates are already in same set
d) Do not remove a randomly chosen wall if the cells it separates are already in same set.
187. If Ahmad is cousin of Ali and Ali is cousin of Asad then Ahmad is also cousin of Asad. This statement has the following property
a) Reflexivity
b) Symmetry
c) Transitivity
d) All of the given
188. A Linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non Linear Data Structure?
a) Arrays
b) Linked Lists
c) Binary Search Trees
d) Stack
189. Recursive call of a function use data structure.
a) Linked List
b) Queue
c) Stack
d) Table
190.is the stack characteristic but.....was implemented because of the size limitation of the array.
✚ isFull (), isEmpty ()
✚ pop (), push ()
✚ isEmpty (), isFull ()
✚ push (), pop ()
191. Which of the following operations returns "most recently entered value" from the stack?
✚ Push
✚ Recent
✚ Top
✚ First
192. Which of the following operations returns "most recently entered value" from the stack?
✚ Reference
✚ Value
✚ Reference & value
✚ Copy of the object can not be passed

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193. The variable which are destroyed automatically when a function's execution ends are:

- ☐ Global variables
- ☒ Local variables defined inside function body
- ☐ Variables (objects) defined inside function body dynamically
- ☐ Variables (objects) defined inside function body statically

194. The.....of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

- a) Height
- ☒ b) Balance
- c) Width
- d) None of the above

195. Which operation of queue data structure is used to get front element from the queue and then remove it from the queue?

- ☒ a) enqueue()
- b) dequeue()
- c) front()
- d) remove()

196. Suppose we have been given the following data set for a Queue.

7 5 2 4

What will be the resultant Queue if we call enqueue(3) method?

Note that 7 is the front element whereas 4 is rear element of queue.

- a) 7 5 2 4
- b) 7 5 2 4
- ☒ c) 7 5 2 4 3
- d) 5 2 4 3

197. In C++, we place the class interface in.....file.

- ☒ a) .cpp
- b) .cppp
- c) .h
- d) .hh

198. The array in binary search is sub divided.....

- a) Once
- b) Twice
- c) N time
- ☒ d) Untill a sublist is no more divisible

199. If unions are done by weight (size), the depth of any element is never greater than

- a) $\log 3n$
- ☒ b) $\log_2 n$
- c) $n \log_2 n$
- d) $\log n*n$

200.in AVL is logarithmic.

- a) Updating
- ☒ b) Searching
- c) Deletion
- d) Insertion

201. What is the depth of any tree if the union operation is performed by height?

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- a) $O(N)$
b) $O(N \log N)$
c) $O(\log N)$
d) $O(M \log N)$
202. Avl tree takes maximum.....time to search an element.
a) $1.44 \log_2 n$
b) $\log_2(n+n)$
c) $\log_2(n+1)+1$
d) $1.88 \log_2 n$
203. Which of the following is correct about AVL Tree?
a) It is identical to BST except height of the left and right subtrees can differ by at least 1.
b) It is identical to BST except height of the left and right subtrees must differ by at least 1.
c) It is not identical to BST, it is totally different kind of tree.
d) It is identical to BST except height of the left and right subtrees can differ by at most 1.
204. "new int[11]" will allocate memory for.....integers.
a) 10
b) 11
c) 12
d) 13
205. The..... method of list data structure removes the element residing at the current position
a) Add
b) Next
c) Remove
d) Find
206. While implementing non-recursive traversal for Binary Search Tree, we need to implement _____.
a) Queue
b) Stack
c) Min heap
d) Max heap
207. Doubly Linked List always has.....NULL pointer/s in a node.
a) One
b) Two
c) Three
d) Four
208. For a perfect binary tree of height h, having N nodes, the sum of heights of nodes is _____.
a) $N - h - 1$
b) $N - 1$
c) $N - 1 + h$
d) $N - (h - 1)$
209. If there are N elements in an array then the number of maximum steps needed to find an element using Binary Search is _____.
a) N

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- b) N^2
- c) $N \log_2 N$
- d) $\log_2 N$

210. Here is a small function definition:

```
void f(int i, int &k)
{
i = 1;
k = 2;
}
```

Suppose that a main program has two integer variables x and y, which are given the value 0. Then the main program calls f(x,y); What are the values of x and y after the function f finishes?

- a) Both x and y are still 0.
- b) x is now 1, but y is still 0.
- c) x is still 0, but y is now 2.
- d) x is now 1, and y is now 2.

211. The worst case of searching in binary search tree (BST) is:

- a) When the data inserted in BST is sorted
- b) When the height of left sub-tree is greater than right sub-tree
- c) When the height of right sub-tree is greater than left sub-tree
- d) When the tree is balanced

212. What is the hash function used in linear probing?

- a) $h_i(x) = \text{hash}(x) \bmod \text{table size}$
- b) $h_i(x) = (\text{hash}(x) + f(i^2)) \bmod \text{table size}$
- c) $h_i(x) = (\text{hash}(x) + f(i)) \bmod \text{table size}$
- d) $h_i(x) = X \bmod 17$

213. Each operator in a postfix expression refers to the previous _____ operand(s)

- a) One
- b) Two
- c) Three
- d) Four

214. In 1990, Bill pugh proposed an enhancement on linked lists and the new data structure was termed as

- a) Linked list
- b) B-Tree
- c) Skip list
- d) Spelling checker

215. Which of the following statement is NOT correct regarding Table ADT?

- a) In a table, the type of information in columns may be different.
- b) A table consists of deveral columns known as entities
- c) The row of a table is called a record
- d) A major use of table is in databases where we build and use tables for keeping information

216. A binary tree of N nodes has.....

- a) $\log_{10} N$ levels
- b) $\log_2 N$ levels
- c) $N / 2$ levels

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d) $N \times 2$ levels

217. If the height of a perfect binary tree is 4. What will be the total number of nodes in it?

a) 15

b) 16

c) 31

d) 32

218. Which property of equivalence relation is satisfied if we say:

Ahmad R(is related to) Ahmad

a) Reflexivity

b) Symmetry

c) Transitivity

d) All of the given

219. If Ahmad is boss of Ehsan and Ehsan is boss of Umer then Ahmad is also boss of Umer. The above-mentioned relation is.....

a) Reflexive

b) Symmetry

c) Transitive

d) None of the given

220. Which of the following statement is false?

a) Arrays are dense lists and static data structure

b) data elements in linked list need not be stored in adjacent space in memory

c) pointers store the next data element of a list

d) linked lists are collection of the nodes that contain information part and next pointer

221. Binary search algorithm cannot be applied to.....

a) Sorted linked list

b) sorted binary trees

c) sorted linear array

d) None of the given

222. Suppose there is an image segmented into pixels. Each pixel has _____ neighbour(s).

a) 0

b) 4

c) 8

d) 16

223. In singly linked list a node consists of two parts:

a) Object and structure

b) Two pointers

c) Two objects

d) Object and pointer

224. A hash function returns avalue.

a) Integer

b) Double

c) Float

d) Char

225. In a tree, we link the nodes in such a way that it.....a linear structure.

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- a) does not remain
b) forms
c) reverses
d) remains
226. The principal benefit of a link list over a conventional array is that the order of the linked items may be from the order that the data seems are stored in memory.
a) Different
b) Identical
c) Same
d) Equivalent
227. The Computer memory can be thought of as a/an.
a) List
b) Queue
c) Stack
d) Array
228. Before using the Pop method of a stack, the user must call the __method.
a) isFull()
b) push()
c) pop()
d) isEmpty()
229. A stack carries behavior.
a) FIFO
b) FEFO
c) LIFO
d) AVCO
230.Method returns the top element of the stack without removing it.
a) Pop()
b) Front()
c) Push()
d) Top()
231. In Queue data structure element are removed from.....
a) Pop
b) Push
c) Rear
d) Front
232. In Queue data structure element are full from____.
a) Pop
b) Push
c) Rear
d) Front
233. “end()” method of list performs its tasks in
a) Many Steps
b) Three Steps
c) One Steps
d) Two Steps
234. Linked List use to store data.
a) Array

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- b) Variables
 - c) Linked Memory
 - d) 2-D Array
235. Complete the push method code of stack void push (int x) { A{____}=x}
- a) Count++
 - b) Count—
 - c) ++count
 - d) —count
236. A template is a Function or Class that is Written adata type.
- a) Generic
 - b) None of These
 - c) Define
 - d) Specific
237. Trying to remove an element from an empty Stack is called____
- a) Garbage Collection
 - b) Overflow stack
 - c) Empty collection
 - d) Underflow of Stack
238. Stack and Queue can be implemented using.....
- a) Binary Tree
 - b) Singly Linked List
 - c) AVL Tree
 - d) Binary Search Tree
239. A software solution is said to be efficient if it solves the problem.
- a) With its resource constraints
 - b) With No Time
 - c) By Consuming more hardware resources.
 - d) By Using some extra resources.
240. we cannot remove items randomly from.....
- a) Stack
 - b) Queue
 - c) Stack and Queue
 - d) List
241. Stack push(15) will push 15 on.....
- a) Top of the Stack
 - b) Anywhere of the Stack
 - c) Middle of the Stack
 - d) Bottom of the Stack
242. Which of the following is the correct conservation from infix to postfix expression $A*B+C/(E-F)$.
- a) $ABC*+EF-/$
 - b) $AB*CFE-/*$
 - c) $AB*C+EF/*$
 - d) $AB+C*E-F/$
243. Convert the Given infix from $12+60-23$ of expression in postfix form.
- a) $12+60\ 23-$
 - b) $12\ 60+ -23$

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- c) -12 60 -23
d) None
244. `Y=&x[0];`
In the above statement, we get the address of the Locatio of the array x and store it in y.
Here “Y” is;
a) Xvalue
b) Zvalue
c) Ivalue
d) Nvalue
245. Last Node in Circular Linked list Contains.
a) Maximum Two points
b) No Null Pointer
c) Atleast one Null pointer
d) None
246. Consider the linked list having data [6, 72, 35, 65,25] stored in it. While current pointer is pointing memory location having 72 stored in it. After calling add(4) function on the following linked list current point will point to memory location having value?
a) 36
b) 4
c) 72
d) 25
247. there is no such Node whose next filed is NULL which one of the given option support the statement
a) Linked List
b) Circular Link list
c) Array
d) Queue
248. Non-recursive calls are faster than which of the following calls?
+ Parameterized
+ Recursive
+ Function
+ Non-function
249. The balance of a node is the result of “height of left subtree””height of right subtree”.
+ Plus
+ Minus
+ Multiply
+ Divided by
250. Searching an element in an AVL tree takes maximum.....time (where n is number of nodes in AVL tree)
+ $\text{Log}(n+1)$
+ $\text{Log}(n+1)-1$
+ $1.44 \log_2 n$
+ $1.66 \log_2 n$
251. A complete binary tree having “N” nodes consists of.....levels.

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- ✚ Log2 (N+1)-1
 - ✚ Log2 (N-1) -1
 - ✚ Log2 (N+1) +1
 - ✚ Log2 (N-1) +1
252. In the post-order traversal of a binary search tree, nodes process as:
- ✚ Left-subtree, Right-subtree, Root
 - ✚ Right-subtree, Root, Left-subtree
 - ✚ Left-subtree, Root, Right-subtree
 - ✚ Right-subtree, Left-subtree, Root
253. The simplest case in a BST to delete a node is:
- ✚ When the node, that is to be deleted is root node
 - ✚ When the node, that is to be deleted has both left and right child
 - ✚ When the node, that is to be deleted has only one child
 - ✚ When the node, that is to be deleted is a leaf node
254. If class A define class B as its friend, then:
- ✚ Class A can access private numbers of class B
 - ✚ Class B can access private numbers of class A
 - ✚ Class A can access only the public members of class B
 - ✚ Class B can access only the public members of class A
255. In the statement `int& a=b;`
- ✚ a and b pointing to two different memory location
 - ✚ a and b are two different names of the same memory location
 - ✚ a and b are two different variables names
 - ✚ b holds the address of variable a
256. In simple implementation of stack, is Full() method is used due to.....
- ✚ Limitation of array
 - ✚ Strength of array
 - ✚ Linked list connectivity
 - ✚ Complexity of linked list
257. Write in postfix from $5*(9-7)$
- ✚ 5(9 7-)*
 - ✚ 5 9 *7-
 - ✚ 5 9 7 - *
 - ✚ 5 (9 7 -)
258. What will be result of following postfix expression?
- 1 2 3*2-
- ✚ 3
 - ✚ 4
 - ✚ 5
 - ✚ 10
259. An array x of 100 integers is declared as,
- ✚ `int x = [100];`
 - ✚ `int x [100];`
 - ✚ `integer array x=100;`

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- integer x [] =100;
260. What is the maximum depth of recursive calls a function may make?
- 1
 - 2
 - n (where n is the argument)
 - There is no fixed maximum
261. A linear Data Structure is the data structure in which data elements are arranged in a sequence or a linear list. Which of the following is Non-Linear Data Structure?
- Arrays
 - Linked lists
 - Binary Search Trees
 - Stack
262. The new operation in C++ for dynamically allocating memory returns,
- Size of the memory it has allocated
 - Pointer to the memory it has allocated
 - Both size and pointer to the memory it has allocated
 - Value of the memory it has allocated
263. In....., a program uses two pointers in the node, i.e one to point to next node and the other to point to the previous node.
- Linked list
 - Doubly-linked list
 - Array
 - Structure
264. The expression $AB+C^*$ is called?
- Prefix expression
 - Postfix expression
 - Infix expression
 - Prefix and infix expression
265. Consider the function X as under
- ```
int X (int& Value)
{
 Return Value;
}
```
- Now a and b are integers in a calling function. Which one of the following is a valid call to the above function X?
- a = X (b);
  - a = X (&b);
  - a = X (\*b);
  - a = X (&\*b);
266. Each node in Binary Search Tree has,
- 1 pointer
  - 2 pointers
  - 3 pointers
  - 4 pointers

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267. Suppose  $n$  is the number of nodes in a complete Binary Tree, then maximum steps required for a search operation are,
- ☒  $\log_2(n+1)-1$
  - ☐  $\log_2(n+1)$
  - ☐  $\log_2(n)-1$
  - ☐  $\log_2(n)$
268. The next field in last node of a singly-linked list is set to.....
- ☐ NAN
  - ☐ 1
  - ☒ NULL
  - ☐ -1
269. Consider the following function:
- ```
voidtest_a(int n)
{
    cout << n << " ";
    if (n>0)
        test_a(n-1);
}
```
- ☐ 4 2 0
 - ☐ 0 2 4
 - ☐ 4 2
 - ☐ 2 4
270. $a*(b + c)-d$ is an example of.....expression.
- ☒ infix
 - ☐ prefix
 - ☐ postfix
 - ☐ allfix
271. Below the binary Search Tree (BST). If we delete the value 50 from the root node, what would be the value in the root of the remaining tree?
- ☐ 50
 - ☐ 60
 - ☐ 70
 - ☒ 80
272. Which of the following calling methods does not change the original value of the arguments in the calling function?
- ☒ Call by passing the value of the arguments
 - ☐ Call by passing reference of the arguments
 - ☐ Call by passing the address of the arguments
 - ☐ Call by passing pointer of the arguments
273. Which of the following statement is NOT true for reference variable?
- ☐ Once a reference is created, it cannot be later made to reference another object.
 - ☐ Reference cannot be NULL
 - ☒ Reference can be uninitialized
 - ☐ It is not possible to refer directly to a reference object after it is defined
274. Which of the following statement is correct?
- ☒ Array size is fixed once it is created

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- ✚ Link List size is fixed once it is created
 - ✚ Binary Search Tree size is fixed once it is created
 - ✚ AVL Tree size is fixed once it is created
275. the data of the problem is of 2GB and the hard disk of 1GB capacity, to solve this problem we should
- ✚ Use better data structures
 - ✚ Increase the hard disk space page 5
 - ✚ Use the better algorithm
 - ✚ Use as much data as we store on the hard disk
276. In call by value methodology, a copy of the object is passed to the called function
- ✚ False
 - ✚ True page 202
277. The tree structure is a
- ✚ Linear data structure
 - ✚ Non-linear data structure page 112
 - ✚ Graphical data structure
 - ✚ Data structure like queue
278. When should you use a const reference parameter?
- ✚ Whenever the parameter has huge size
 - ✚ Whenever the parameter has huge size, the function changes the parameter within its body, and you do not want these changes to alter the actual argument.
 - ✚ Whenever the parameter has huge size, the function changes the parameter within its body, and you do want these changes to alter the actual argument.
 - ✚ Whenever the parameter has huge size, the function changes the parameter within its body
279. Here is the start of a C++ class declaration:
- ```
class foo
{
 public:
 void x(foo f);
 void y(const foo f);
 void z(foo f) const;
```
- Which of three members function can alter the PRIVATE member variables of the foo object that activates the function?
- ✚ Only x can alter the private member variables of the object that activates the function
  - ✚ Only y can alter the private member variables of the object that activates the function
  - ✚ Only z can alter the private member variables of the object that activates the function
  - ✚ Two of the functions can alter the private member variables of the object that activates the function

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280. suppose we have a circular array implementation of the queue class, with ten items in the queue stored at data[2] through data[11]. The CAPACITY is 42, i.e the array has been declared to be of size 42. Where does push member function place new entry in the array.

- ☐ data [1]
- ☐ data [2]
- ☐ data [11]
- ☒ data [12]

281. ....is a binary tree where every node has a value, every node's left subtree contains only values less than equal to the node's value, and every node's right subtree contains only values that are greater than or equal?

- ☐ Strictly Binary Tree
- ☒ Binary Search Tree
- ☐ AVL Tree
- ☐ All of these

282. We access elements in AVL tree in,

- ☐ Linear was only
- ☒ Non-linear was only
- ☐ Both linear and non-linear ways
- ☐ None of the given options

283. AVL tree is,

- ☒ Non-linear data structure
- ☐ Linear data structure
- ☐ Hybrid data structure (Mixture of linear and non-linear)
- ☐ None of the given options

284. Each operator in a postfix expression refers to the previous.....operand(s).

- ☐ One
- ☒ Two
- ☐ Three
- ☐ Four

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285. Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What boolean expression will be true when cursor refers to the tail node of the list?

- ☒ (currentNode == null)
- ☐ (currentNode->nextNode == null)
- ☐ (nextNode, data == null)
- ☐ (currentNode, data == 0.0)

286. Suppose that the class declaration of SomeClass includes the following function prototype.

Bool Less Than (SomeClass anotherObject);

Which of the following tests in the client code correctly compares two class objects alpha and beta?

- ☐ If (alpha < beta)
- ☒ If (alpha.LessThan (beta))
- ☐ If (LessThan(alpha,beta))
- ☐ If (LessThan(alpha). beta )

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287. In C what is the operation that you can not do with primitive types?
- ✚ Assign a value to primitive type using a literal
  - ✚ Declare primitive types to be constant using the Const keyword
  - ✚ Create new instance of primitive type with new Keyword
  - ✚ None of these
288. The operations for adding an entry to a stack is traditionally called:
- ✚ add
  - ✚ append
  - ✚ insert
  - ✚ push page 53
289. The operation for removing an entry from a stack is traditionally called:
- ✚ delete
  - ✚ peek
  - ✚ pop
  - ✚ remove
290. Consider the following sequence of push operations in stack:
- ```
stack.push('7');  
stack.push('8');  
stack.push('9');  
stack.push('10');  
stack.push('11');  
stack.push('12');
```
- ✚ 7 8 9 10 11 12
 - ✚ 9 8 11 10 7 12
 - ✚ 9 10 8 11 12 7
 - ✚ 9 10 8 12 7 11
291.is the maximum number of nodes that you can have on stack-linked list?
- ✚ Zero
 - ✚ $2n$ (where n is the number of nodes in linked list)
 - ✚ Any number
 - ✚ None of these
292. Which of the following can be used to reserve a string value,
- ✚ Stack
 - ✚ Queue
 - ✚ Both of these
 - ✚ None of these
293. The following are statement related to queues.
- (i) The last item to be added to a queue is the first item to be removed
 - (ii) A queue is a structure in which both ends are not used
 - (iii) The last element hasn't to wait until all elements preceding it on the queue are removed
 - (iv) A queue is said to be a last-in-first-out list or LIFO data structure.
- Which of the above is/are related to normal queues?
- ✚ (iii) and (ii) only
 - ✚ (i), (ii) and (iv) only
 - ✚ (ii) and (iv) only
 - ✚ None of these

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294. An array is a group of consecutive related memory locations.
☒ True
☐ False
295. In an array we can store data elements of different types.
☐ True
☒ False page 7
296. In an array list the current element is
☒ The first element
☐ The middle element
☐ The last element
☐ The element where the current pointer points to
297. Which of the following calling methods does not change the original value of the argument in the calling function?
☐ None of the given
☒ Call by passing the value of the argument
☐ Call by passing reference of the argument
☐ Call by passing the address of the argument
298. Which of the following statement is NOT correct?
☒ Array size can be changed after its creation
☐ Link list size can be changed after its creation
☐ Binary Search Tree size can be changed after its creation
☐ AVL Tree size can be changed after its creation
299. A queue is a.....data structure, whereas a stack is a.....data structure.
☒ FIFO, LIFO page 161,54
☐ LIFO, FIFO
☐ None of these
☐ Both (a) and (b)
300. Which of the following operators has higher priority than all of others?
☒ Multiplication operator
☐ Minus operator
☐ Plus, operator
☐ Exponentiation operator
301. Four statements about trees are below. Three of them are correct. Which one is incorrect?
☐ Trees are recursively defined multi-dimensional data structures tree
☐ The order of a tree indicates a maximum number of children allowed at each node of the
☐ A search tree is a special type of tree where all values (i.e. keys) are ordered
☒ If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height
302. Which of the following is "TRUE" about arrays,
☐ We can increase the size of arrays after their creation.
☐ We can decrease the size of arrays after their creation.
☐ We can increase but can't decrease the size of arrays after their creation.
☒ We can neither increase nor decrease the size of arrays after their creation.
303. There is/are case/s for rotation in an AVL tree,
☒ 1

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✚ 3

✚ 2

✚ 4

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304. Consider the following statements.

- (i) A binary tree can contain at least 2_L nodes at level L .
- (ii) A complete binary tree of depth d is a binary tree that contains 2_L nodes at level L between 0 and d , both inclusive
- (iii) The total number of nodes (T_n) in a complete binary tree of depth d is $2_{d+1} - 1$.
- (iv) The height of the complete binary tree can be written as $h = \log_2 (T_n + 1) - 1$ where T_n is total number of nodes.

Which of the following is correct in respect of the above statements regarding the Binary Trees?

✚ (i) and (iii) only

✚ (i), (ii) and (iii) only

✚ (ii) and (iii) only

✚ (ii), (iii) and (iv) only

305. Consider the following infix expression.

$5 + 6/2$

If one converts the above expression into postfix, what would be the resultant expression?

✚ $56/+2$

✚ $5\ 6\ 2/+$

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✚ $5\ 6/2+$

✚ $/62 + 5$

306. Addition of new items in stack make the pointer.....by 2

✚ Increment, bits

✚ Increment, bytes

✚ Decrement, bits

✚ Decrement, bytes

307. Next item in a linked list is known as

✚ Index

✚ Item

✚ Node

✚ Child

308. In an AVL tree to delete a parent with two childs in a straight-line following rotation will be required: -

✚ Single

✚ Double

✚ Triple

✚ None

309. To check the depth of an AVL tree following time will be taken: -

✚ $1.66 \log_{2n}$

✚ $1.44 \log_{2n}$

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✚ $\log_2(n+1) - 1$

✚ $1.66 \log_{2n} (n+1)$

310. After creation of an array: -

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- ☐ Size can be increase but cannot be decreased
 - ☐ Size can be decrease but cannot be increased
 - ☒ Size can be neither increase nor be decreased
 - ☐ Size can be increase and can also be decreased
311. Each node in a BST has Pointers: -
- ☐ 1
 - ☒ 2
 - ☐ 3
 - ☐ 4
312. Highest Operators Precedence is of the following operator: -
- ☐ Plus
 - ☐ Minus
 - ☒ Multiply
 - ☐ Exponentiation
313. Following is linear data structure: -
- ☐ Stacks
 - ☐ Queues
 - ☒ Both a & b page 52,87
 - ☐ None of these
314. Each entry which points to a null value in a Singly Linked List is known as: -
- ☐ Node
 - ☐ First nide
 - ☒ Last node
 - ☐ Head node
315. Non-recursive calls are faster than the Recursive calls: -
- ☒ True page 323
 - ☐ False
316. The data structure is a
- ☐ Linear
 - ☒ Non-linear page 112
 - ☐ Circular
 - ☐ None of above
317. What will be the valid postfix notation of $A+B*C-D$
- ☐ $ABC+*D$
 - ☒ $ABC*+D-$
 - ☐ $ABCD+-*$
 - ☐ $AB+D*C$
318. When an operator is used in between two operands this is which type of notation
- ☐ Prefix
 - ☐ Postfix
 - ☒ Infix page 64
 - ☐ None of these
319. A compound Data Structure is the data structure which can have multiple data items of same type or of different types. Which of the following can be considered compound data structure?
- ☒ Arrays
 - ☐ Link lists

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- ☐ Binary Search Trees
 - ☐ All of the given
320. Suppose a pointer has been declared in main but has not assigned any variable address then
- ☐ That pointer points to First byte in main function
 - ☒ That pointer points contains a NULL value
 - ☐ None of these
 - ☐ That pointer points to any memory address
321. Which statement of the following statements is incorrect?
- ☐ Lists can be implemented by using arrays or linked lists
 - ☐ A list is a sequence of one or more data items
 - ☐ Stack is a special kind of list in which all insertions and deletions take place at one end
 - ☐ Stacks are easier to implement than lists
322. Consider the following function:
- ```
void test_a(int n)
{
 cout << n << " ";
 if (n>0)
 test_a(n-2);
}
```
- What is printed by the call test\_a(4)?
- ☒ 4 2 0
  - ☐ 0 2 4
  - ☐ 0 2
  - ☐ 2 4
323. Queue follows,
- ☐ Last in First out
  - ☐ First in Last out
  - ☒ First in First out
  - ☐ None of these
324. In a data structure that can grow easily dynamically at run time without having to copy existing elements?
- ☐ Array
  - ☒ List
  - ☐ Both of these
  - ☐ None of these
325. Which one of the following statements is NOT correct.
- ☒ In linked list the elements are necessary to be contiguous
  - ☐ In linked list the elements may locate at far positions in the memory
  - ☐ In linked list each element also has the address of the next to it
  - ☐ In an array the elements are contiguous
326. linked lists are collections of data items "lined up in a row" , insertions and deletions can be made only at the front and the back of a linked list.
- ☐ Arrays
  - ☐ Link lists

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## ☒ Binary Search Trees

- ☒ None of these
327. Which one of the following is correct about pointers?
- ☒ They always point to different memory locations
  - ☒ They always point to a single memory location
  - ☒ The address of two pointer variables is same
  - ☒ None of these
328. Which of the following abstract data types are NOT used by Integer Abstract Data type group?
- ☒ short
  - ☒ int
  - ☒ float
  - ☒ long
329. We can add elements in QUEUE from.....
- ☒ Front
  - ☒ Rear
  - ☒ From both rare and front
  - ☒ None of these
330. The following is a segment of a C program.
- ```
int pqr(BinaryNode t)
{if (t == null)
    return -1;
else
    return 1+max(pqr(t.left), pqr(t.right)) }
```
- Identify, what the above program intends to do?
- ☒ Compute the height of a binary tree using in-order traversal
 - ☒ Compute the height of a binary tree using pre-order traversal
 - ☒ Compute the depth of a binary tree using pre-order traversal
 - ☒ Compute the depth of a binary tree using post-order traversal
331. Consider the following infix expression:
- $$3 + 5 * 6 - 7 * (8 + 5)$$
- Which of the following is a correct equivalent expression(s) for the above?
- ☒ $3 * 6 * 5 + 7 * 5 * 8 + - *$
 - ☒ $3 * 6 * 5 * 7 * 5 * 8 + * + - *$
 - ☒ $3 * 5 * 6 + * 7 * 8 * 5 + - *$
 - ☒ $3 * 5 * 6 * + 7 * 8 * 5 + * -$
332. Is this a correct statement? Give answer Yes or No.
- A node cannot be deleted , when the node to be deleted has both left and subtrees.
- ☒ Yes
 - ☒ No
333. Deleting a leaf node in binary search tree involves setting.....pointers of that node's parent as null.
- ☒ 1
 - ☒ 2
 - ☒ 3
 - ☒ 4

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